

MIT 18.642 · 22 VIDEO TRANSCRIPTS · COMPREHENSIVE NOTES

Ghi chú đầy đủ theo transcript cho Hòa Đại ka

Bám sát thứ tự nội dung video: mục tiêu, chủ đề, diễn giải, timeline, câu transcript tiêu biểu, điểm cần nhớ.

Mục lục 22 video

01. Lecture 1, Part I: Introduction of the Class

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02. Lecture 1, Part II: Introduction of Financial Markets, Financial Terms and Concepts

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81,640 ký tự transcript

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16. Lecture 18: Applying Data Science and Artificial Intelligence to Managing Biomedical Portfolios

69,137 ký tự transcript

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18. Lecture 21: Black-Scholes Formula, Risk Neutral Valuation

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19. Lecture 20: Building the First Federally (CFTC) Regulated Exchange Dedicated to Trading on Events

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21. Lecture 24: Stochastic Calculus

39,155 ký tự transcript

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42,387 ký tự transcript

Lecture 1, Part I: Introduction of the Class

Source transcript: mit_18_642_transcripts\01_b8u2CQLQBVU.json

1. Chủ đề nhận diện trong video

event markets

machine learning

black-scholes/options

portfolio/risk

rates/bonds

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Trọng tâm liên quan lãi suất/trái phiếu: chiết khấu, yield, đường cong lãi suất, duration hoặc sản phẩm rates. Đây là nền cho định giá dòng tiền và hiểu môi trường lãi suất. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:11 — VASILY STRELA: So today's class is introductory class. We'll talk about the class itself and about the program of the class. Then Jake will talk a little bit about the financial terms and ideas. And I'll talk about bond math a little bit. So the class, because it is organized to introduce you to the applications of math in the real world, the class is split between a mathematical part, which will be taught by Peter, and application part, which will be taught by many...

05:39 — about applying some techniques from computational finance into biomedicine and biomedical portfolio. Oh, I missed both Jake and myself there, but Jake will talk about portfolio optimization, and I'll talk about Black-Scholes. And the last invited lecture will be by John Hull, who is legendary in derivatives space, but he will actually talk about machine learning, his newest interest. With that, let me pass it on to Peter. PETER: OK, thank you, Vasily. Let me just page back up to the schedule of lectures. Although,...

10:15 — We will also be using the language R in our studio for illustrations. And I think most students here probably are Python experts. I am not a Python expert, so I use R. But let me just tab over to a slide showing this. Let's see. Using R, one of the reasons for using R is that it gives you immediate access to state-of-the-art statistical methods that get developed. But in addition to that, it facilitates the downloading and collection of financial data from the...

15:19 — And this was an event that the brokers, important brokers, didn't program in that possibility into their systems. And so when the system became updated on this day when the price was significantly negative, those who held, say, thousands of contracts of the contract actually owed money to the exchange for their negative position. Anyway, the financial markets offer these cases where surprising, seemingly impossible events can occur, such as negative interest rates. When I was studying finance, the thought of negative interest rates was...

4. Câu/ý tiêu biểu từ transcript

“I taught in the statistics department at Harvard and then joined the Sloan School, where I pursued research on statistical modeling and finance.”

“But one can see issues like nonlinear volatility models, regime modeling those are packages that are quite useful and powerful.”

“Then Stefan Andreev from Two Sigma will talk about principal component analysis and finance, which should go well together with what Peter will be talking about at the time.”

“Then James Sheppard from Quantile Technologies, which is now part of London Stock Exchange, will talk about optimization and compressing derivative portfolios.”

“Then Andrew Gustavsson from Mizuho will talk about linear rates, product swaps, bonds, and curve construction.”

“Oh, I missed both Jake and myself there, but Jake will talk about portfolio optimization, and I'll talk about Black-Scholes.”

“But here, were two students who basically create an exchange where you bet on discrete events, whether things will or will not occur.”

“Andrew Lo from MIT will talk actually about applying some techniques from computational finance into biomedicine and biomedical portfolio.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 1, Part II: Introduction of Financial Markets, Financial Terms and Concepts

Source transcript: mit_18_642_transcripts\02_z4p87TPCnQc.json

1. Chủ đề nhận diện trong video

machine learning

black-scholes/options

rates/bonds

portfolio/risk

event markets

2. Giảng lại đầy đủ ý chính

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Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:12 — JAKE XIA: So in my part of the first lecture, I will, before I get into the introduction of some financial concepts and investment strategies, give you some basics on the background information for this class. I will tell you three things first before that. First is a bit of introduction of myself. Second, I tell you how this class was actually formed, why we had this idea, why we decided to have this class. Third is, before you actually do the assignment 0, submit...

04:30 — to answer all the questions or get to know the students. So we figured maybe the best way is to have a seminar to tell the students more about how math is used in finance. So one seminar lecture actually got expanded to the whole term. And so we invited different practitioners talking about how methods being applied in different fields, solving different problems that the class actually got very popular. I mean, one of the first classes, we've got students sit on the floor...

08:34 — I need something. So let's exchange. So the Stock Exchange was formed with the same idea. So some buyers want to own equity of a company. I'll explain what equity is. Equity is basically ownership of a company, meaning stocks. And they centralize all the trading of these stocks at a particular place. These are exchanges. Many cities, large financial centers, have different exchanges. And in the last few decades, that also has changed. It's no longer that centralized. So people do a lot of...

13:24 — But they can also default on that payment, on both the principal or the interest. So bond has a credit risk, meaning that the issuer may not always pay back. So that involves the exposure of the credit of the issuer. So that's basically related to a whole bunch of credit related products. And 2008, global financial crisis is really a credit crisis because that started in the mortgage market. People borrowed money to buy houses, but couldn't pay back. And also, these mortgages were...

17:31 — Bank of America, Chase. They take deposits. They also lend out the money to home builders or other commercial projects. They are the investment banks where you

interview. They come to MIT to recruit-- so the Morgan Stanley's, the Goldman Sachs, and others, JP Morgan's. JP Morgan was actually a commercial bank. But since the termination of Glass-Steagall law in 1999, they turned themselves into an investment bank-- they turned themselves back into investment bank because, many years ago, before 1930, Morgan Stanley and JP...

22:00 — If you one day want to raise a hedge fund, you need to deal with your investors. So you need to sort out these kind of relationships. So the trader types we can categorize into the three. One is the hedgers, like corporates. They want to reduce risk. And the second is the market maker. Like a lot of the bank trading desks, they make a buying price, a selling price, or bid, or an offer. Their objective is to earn that spread, but not...

25:56 — And I would say some of them are very successful. Some of the funds out there, they have achieved amazing outcomes using mathematics. But the problem is, once they found those secrets, they would never disclose it. And if you join them, they will ask you to sign a lifetime non-compete, meaning, when you leave the shop, you cannot say a word of what you learned. Yeah, go ahead. STUDENT: If you sign a lifetime non-compete, don't they have to pay settlements when they fire...

29:46 — Those \$100 bills have definitely been picked up. But sometimes, they still show up. Do you want to train following or mean reversion? That means, when market goes up, do you want to buy more? Or market goes up, do you want to sell? You play the contrarian or you buy the dip. Do you want to focus on short term or you focus on long term? Short term, you're trading very fast and you're trading the liquid. You can change your mind easily. Or...

33:41 — So you need to track on the spreadsheet how many shares you have bought. That's easy. \$10,000, right? You just convert into shares divided by per share price. And your daily profit and loss in dollar and percentage, that gives you a practice of how to track your position. Every day, there's a P&L, profit and loss. At the end of the close on November 15, calculate the total loss or profit. And also, I want you to do one calculation to sum up all...

4. Câu/ý tiêu biểu từ transcript

“One is to come up with a pricing model, because these instruments, especially the derivatives, the options, the forwards, the swaps is actually quite complicated.”

“You can also join the so-called IBD, investment banking division, which is more about corporate finance or raising money, taking a company from private to be a listed, public company or issuing bonds or some other ways.”

“So they have to go through a process called IPO, initial public offering, to become listed on the exchange.”

“But Salomon Brothers, back then, was the pioneer of using quantitative trading strategies and also hiring mathematicians, of doing those so-called arbitrage models, finding one stock or bond versus another, basically finding relative value.”

“They can take the risk So the trade between the lender and the borrower is essentially the main driving force of all the financial relationships.”

“So you probably heard the Nobel Prize given in the '90s to the Black-Scholes-Merton model.”

“So this game is about, for every single one of you, to pick a stock or ETF, exchange traded funds, just traded like stock, or bonds, or whatever only to buy.”

“So the exchange traded stocks is called the listed stocks.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.

- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 1, Part III: Bond “Mathematics”

Source transcript: mit_18_642_transcripts\03_NZ3Mva95UsQ.json

1. Chủ đề nhận diện trong video

rates/bonds

black-scholes/options

regression

machine learning

portfolio/risk

2. Giảng lại đầy đủ ý chính

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Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:18 — VASILY STRELA: The rest of the class, let's talk a little bit about interest rates and a little bit about bonds, because that's very important and central for many-- will be definitely part of many classes and is generally central for finance, and for life, in my opinion. So what is interest? Well, let's start with now. And suppose we have \$1 which we invested-- well, put it in the bank. So what happens in one year? Well, you expect to get your dollar back...

05:33 — Let's borrow \$1-- let's borrow x , this amount now, and let's see what happens at the end. So at the end, in one year, I mean, we'll receive \$1 for sure. And we'll have to repay our debt but with interest. So we'll have repay 1 plus x times r . Sorry-- x times 1 plus. Now, what should this be? Can it be greater than zero, this difference? Well, if it's greater than zero, why wouldn't we borrow-- why wouldn't we buy a right to...

10:48 — at the end, the whole big notional at the end. And can we sum this up? Well, sure. That's geometric sequence. Everybody can sum up geometric sequence. And that's the value. That's what you get. And that's the value of the bond. Now, the bonds are traded on the markets. There are prices of the bonds. And if you start investing into bonds, and not in just in one bond, you should be able to compare to bonds. And comparing the price is not really...

15:38 — because generally speaking, people would like to receive more yield for a longer term investments. And that's what was the case in-- this is what-- this is '21. This is '14. Yeah, nice looking curves. The only difference here is that now they are-- in '92, they were grounded at 3% and went all the way up to 7%, those were grounded at zero and went only to 2%, 3%. Now, interesting observation here is-- let's look at this inverted curves. A year ago it...

20:31 — We will be talking about more interesting interest rate objects in the future, in particular, forward rates and swaps. But I'll leave it to Stefan and Andrew to talk more about that. That's probably all for today. And any questions to any of us? AUDIENCE: Have you done this-- have you done that game before? SPEAKER: Not specifically in this

format. So we tried-- before it was more open-ended. I was thinking-- so I think by fixing the dates, you don't have to make the...

4. Câu/ý tiêu biểu từ transcript

"And this is called discounting, meaning that if we know that we'll receive a certain amount in the future, and we know the interest rates, prevailing interest rates, then we have to discount the future value to today using a discount factor."

"So what's important to know here is that the higher the coupon-- and that's natural-- the higher the yield is for the same bond of the same maturity."

"So what turns out to be more convenient measure to compare bonds and various fixed income securities is yield."

"So very important measures are duration of the bond, which is derivative with respect to yield."

"So for our zero-coupon bond, yield is pretty simple."

"For coupon bonds, yield computation is more complex because inverting this formula is more involved."

"We will be talking about more interesting interest rate objects in the future, in particular, forward rates and swaps."

"So you back out one flat interest rate, which will give you the price of the bond."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 2: Linear Algebra

Source transcript: mit_18_642_transcripts\04_OuimNNluUyY.json

1. Chủ đề nhận diện trong video

linear algebra

portfolio/risk

machine learning

black-scholes/options

pca

2. Giảng lại đầy đủ ý chính

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Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:10 — PETER KEMPTHORNE: All right. Well, welcome to the first math lecture. The topic today is linear algebra. And what I expect is that, for most of you, the topics will be very familiar. At least we'll be covering basic fundamentals of vector algebra, then dealing with special matrices, matrices that are used in applications, eigenvectors and eigenvalue, decompositions, singular value decompositions, and then, finally, the Perron-Frobenius theorem. And what's really neat about quantitative finance is that linear algebra can really be very useful in understanding...

11:00 — And we can calculate the profit and loss of that d portfolio by subtracting the profit and loss of the q and w. Now, let's see, with short selling in markets, let's see-- who in the class might offer a description of what it means to sell short a stock? So suppose that we have $d_{t, \text{sub } 2}$ is equal to, say, negative 200. What would that mean? Or can we actually realize a portfolio where we're short one of the stocks by 200 shares?...

21:43 — a rectangular matrix. And one very useful representation of a matrix is as a collection of column vectors that are stacked one after the other. And so we can denote bold a sub 1 through bold a sub n as the column vectors. And with matrix algebra, let's see, well, we can multiply a matrix, capital A, by a constant. And we end up multiplying each of the components, elements of the table of A values by the same constant c. With a transpose operation,...

33:43 — Then, if we want to compute the probabilities of states at time t plus 1, then it turns out to be simply the matrix vector product of A times π of t. So this formula gives us the model for how the probabilities of states occurring changes from time t to t plus 1. Now, if we consider what is the probability vector at time t plus 2, well, we have the same matrix vector model where the stationary transition probabilities A are multiplied by...

45:27 — T as the percent return. And let's see. We can also talk about the simple rate of return, which would be given by-- well, let's see. Actually, maybe, here I should write R sub B and here R sub B instead of R sub F because I want to define R sub f to be equal

to basically 1 over t , the average rate of return. So it's equal to this times B_T minus B_0 over B_0 . And so this, simply, in this case, is...

59:46 — So we could consider 0 in the down market. And basically, maybe this would be CT minus K here. So this vector here would correspond to the payoff of this call option. So under what circumstances can we take a linear combination of these two vectors and get this vector? Yes? STUDENT: So you've got to short one of the stocks? PETER KEMPTHORNE: Yeah. I mean, basically, we can solve for this here by substituting in for this 0 and P minus K , and then...

71:12 — measure exists on scenarios. And so think about Q star being a probability distribution over the underlying states, ω_1 through ω_m . And it turns out that if the initial price of the j -th asset is the discounted expected terminal price of that j -th asset across the different states, then Q star is called a pricing measure. So our initial prices equal a discount factor times the expected terminal time t values. If such a pricing measure exists, then there's no arbitrage. So it's...

4. Câu/ý tiêu biểu từ transcript

“Now, if we consider what is the probability vector at time t plus 2 , well, we have the same matrix vector model where the stationary transition probabilities A are multiplied by the vector of probabilities at t plus 1 .”

“And so we basically get the square of the transition probability matrix times the π vector at time t .”

“And so if you think of the matrix A having eigenvalues and eigenvectors, what's the eigenvalue and what's the eigenvector in this case?”

“When we have square matrices A , we can consider the existence of an eigenvalue λ and an eigenvector V such that the matrix A times v is equal to just the scalar λ times v .”

“The next topic in our remaining few minutes is eigenvalues and eigenvectors.”

“It turns out that you can prove that a Markov chain is acyclic if all of its components of the A matrix, if every single one, is nonzero, so that there's always a possibility of transitioning from one state to another.”

“But this formula for the stationary distribution is a formula that's very specific to eigenvalues and eigenvectors.”

“And when we have a matrix A that has independent eigenvalues or eigenvectors, then, in this special case, we can consider defining the matrix S to be the matrix whose columns are the different eigenvectors.”

5. Điểm cần ghi nhớ

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- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 4: Linear Algebra (cont.); Probability Theory

Source transcript: mit_18_642_transcripts\05_mtXTs2U1uMA.json

1. Chủ đề nhận diện trong video

linear algebra

probability

portfolio/risk

pca

black-scholes/options

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần đại số tuyến tính giúp biểu diễn tài sản/danh mục/dữ liệu bằng vector và ma trận, là nền cho PCA, covariance và tối ưu danh mục. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. PCA giúp tìm factor chính và giảm trùng lặp dữ liệu, rất hữu ích khi nhiều indicator kể cùng một câu chuyện. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:12 — PETER KEMPTHORNE: All right. I guess we'll get started. Let's see. I wanted to just finish up some discussion of topics from linear algebra that we didn't get to the last math lecture. And to begin with, I think we did cover this topic of eigenvalues and eigenvectors. These linear algebra concepts turn out to be very, very useful in a number of different applications. But when we have an eigenvalue and eigenvector pair, then, for a matrix A that has an eigenvector V , then...

11:34 — So let me write this, maybe, as the singular value decomposition in my notation. So who would like to explain what this is? Anybody? Yes? AUDIENCE: Expresses the matrix A as a sum of orthogonal matrices. PETER KEMPTHORNE: So u is such that $u^T u$ is equal to, say, the identity that has orthogonal columns. V is the same thing. And D is a diagonal. And so what's important is that any matrix can be expressed as a factor of a V^T , which...

23:40 — the value of such a portfolio, where, in fact, as indicated here, it's investing \$1,000 at the beginning of 2019. And as an alternative to just equally weighting across the S&P 500 stocks, I looked at the Amazon, Apple, NVIDIA-- or no, Netflix and Google. These were actually the really hot stocks about four or five years ago. And basically, look at equal weighting of all these. And so here's how the investment, equal weighted at the beginning of the period, resulted towards the end....

34:51 — Basically, the mean of random variables, the probability-weighted outcome, we can generalize the mean to looking at, which is the first moment, to consider the probability-weighted average of the k -th power, where k is an integer. And the variance of random variables is a useful measure of uncertainty about the mean value. And there's a nice formula for the variance, which is the squared distance from the mean in terms of the average squared minus the square of the average. So these are useful formulas...

45:52 — So with this formula, it's all familiar. But if we have f of x is equal to the integral minus infinity to x of f of t dt , then the derivative of f evaluated at x turns out to be equal to f of t evaluated at t equal to x . So the cumulative integral has a derivative equal to the argument of the integral. So this is easy. And we can use this to get the density function for a lognormal distribution. And here's...

57:37 — and the moment-generating functions that can be thought of as a sequence converges to a function that's a moment-generating function of some distribution, then we have that the sequence of random variables, X_s , converges to that distribution as well. So these are nice properties that allow us to understand limits of random variables and what the distributions are of limits. Now, with moment-generating functions, this slide here discusses-- well, it introduces you to the moment-generating function for the normal distribution with mean μ and variance...

69:50 — a linear combination of the X_s given by a vector A of constants, then this linear combination of the X_s can actually represent maybe a portfolio of holdings across different assets given by the X_i 's. And what we'd like to be able to do is model the mean value of Y and its variance. The mean value is very simply the linear combination of the component means times the a coefficients. And when we go to calculate the variance of Y , if you go through...

4. Câu/ý tiêu biểu từ transcript

“And so, with this framework, where we have the covariance matrix, σ , having eigenvalues, λ_i , and corresponding eigenvectors, γ_i , we can normalize the eigenvectors, γ_i , to be of unit length.”

“And with principal components analysis, we consider an m -variate vector x -- so we consider an x vector that's m by 1 , which has an expected value equal to an α vector, and a covariance matrix, which is m by m , equal to σ matrix.”

“And it's also going to have nonnegative eigenvalues because the covariance matrix is positive semidefinite.”

“Some of them will be nonzero, so long as there's no-- so long as the covariance matrix has some positive rank.”

“So we have that the variance of Y is simply the sum, double sum, over all pairs of coefficients, a_i, a_j , multiplied by the covariance terms between i and j .”

“Now, we can compute covariances between two random variables and correlations.”

“So when we have multidimensional vectors that have variability that is limited to a smaller number of dimensions, we can characterize those smaller number of dimensions nicely with principal components and obtain useful models.”

“So this is a nice way of representing the covariance matrix.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 5: Probability Theory (cont.); Stochastic Processes I

Source transcript: mit_18_642_transcripts\06_wMGEKMHsOKE.json

1. Chủ đề nhận diện trong video

probability

portfolio/risk

linear algebra

pca

volatility

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần đại số tuyến tính giúp biểu diễn tài sản/danh mục/dữ liệu bằng vector và ma trận, là nền cho PCA, covariance và tối ưu danh mục. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. PCA giúp tìm factor chính và giảm trùng lặp dữ liệu, rất hữu ích khi nhiều indicator kể cùng một câu chuyện. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Volatility modeling cho thấy biến động thay đổi theo thời gian và ảnh hưởng trực tiếp đến risk, sizing và option/CW.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:12 — PROFESSOR: All right, we'll get started. We're going to continue our discussion of principal components analysis. And what I want to emphasize with principal components is that we're dealing with a random vector X that's in m dimensional space. And it has an expected value of α , say. And it has a covariance matrix given by σ , which is m by m . This kind of setup can be used for modeling returns on stocks or assets that may have a mean return over some holding...

11:57 — or let's see. Or the-- yeah, the covariance matrix of X , in terms of the diagonal entries which represent the correlations, are all due to common loadings on these different principal component variables in p_1 . So the interesting question to think about is, how can we model random vectors with some covariance matrix? And are there simple structures to the covariance matrix that explain all of the intercorrelations? Now, when we perform empirical principal components analysis, then what we do is we-- say, we have...

23:18 — let's see-- arise, especially in the study of regression models and the significance of regression parameter estimates. But basically, with a t distribution, if we take a standard normal and then divide it by the square root of a chi-squared divided by its degrees of freedom, where the chi-squared variable U is independent, then this will give us a t distribution with r degrees of freedom. And so, let's see. With t distributions, t distributions are examples of distributions that are heavier tailed than a...

34:21 — of Z_n , we're going to show that the moment-generating function of Z_n converges to this moment-generating function, which in fact is the normal with mean zero and variance σ^2 . So the theorem that says moment-generating functions are of unique, and if the limit of moment-generating functions exists and equals some function, we can identify the distribution by that limiting moment-generating function. Well, if we calculate just the moment-generating function of Z_n , if we just plug-in Z_n within the expectation, then noting that the X 's...

46:25 — Now, what we want to do is consider what is the optimal choice of X -- the number of shares to buy for this agent? And in order to make that choice, one can apply the decision theory under uncertainty and associate a utility of the end period wealth given by some utility function, u . And we can choose the X value to maximize the expected utility of wealth. So importantly, If we have u of w as a function of w , and there's basically a...

58:38 — of the expected price plus this other term here. And this other term here-- the second derivative is negative. First derivative is positive. So this is actually a negative factor here. And so we end up getting that-- we have a formula that depends upon that ratio of second derivative expectation to first derivative expectation of the utility function. Now, in economics classes, maybe some of you have come across the constant absolute risk aversion utility function. Has anybody come across that in an economics...

71:51 — The capital asset pricing model was actually formulated by John Lintner and William Sharpe back in the early '60s. And I distributed one of the articles of Lintner in the Canvas site. And it's actually nice to be able to read a very technical paper in an accessible way. And so I think that you'll enjoy checking that paper out. OK. Well, let me turn then to the main topic in our syllabus for today, but it's the tail end of today, to talk about...

4. Câu/ý tiêu biểu từ transcript

"The first component corresponds to using this gamma vector, which happens to be an eigenvector of the covariance matrix corresponding to where with associated eigenvalue λ_i ."

"And we then can perform an eigenvalue eigenvector decomposition of this covariance matrix, and then use these results to define the empirical principal component coordinates of our data."

"So with these transformations, as noted here, these principal component variables have basically a mean that's zero and a covariance matrix that's diagonal."

"So this covariance structure it's actually useful to think of a special example, which is the multivariate normal distribution in m dimensions with mean vector α and covariance matrix σ ."

"So the interesting question to think about is, how can we model random vectors with some covariance matrix?"

"Or the-- yeah, the covariance matrix of X , in terms of the diagonal entries which represent the correlations, are all due to common loadings on these different principal component variables in p_1 ."

"And importantly, if this were an appropriate model for an m vector x , then the covariance matrix of the-- the covariance matrix of X is just given by-- or let's see."

"This kind of setup can be used for modeling returns on stocks or assets that may have a mean return over some holding period and a variance-covariance matrix."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 6: Stochastic Processes I (cont.); Regression Analysis

Source transcript: mit_18_642_transcripts\07_yln8Y_CSwpK.json

1. Chủ đề nhận diện trong video

probability

stochastic processes

event markets

machine learning

black-scholes/options

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần quá trình ngẫu nhiên xem giá là đường đi theo thời gian, nên drawdown, hitting time và thứ tự dữ liệu rất quan trọng. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:16 — PETER KEMPTHORNE: All right, we'll get started. See, last time we introduced you to the definition of martingales. And what we're going to do the first part today is to see how useful and powerful martingale properties are and solving interesting questions and challenges. So just as a reminder, if we have a stochastic process x_n , which is just a sequence of random variables, generally the index n corresponds to time. Then we can consider a derived process capital $M_{\text{sub } n}$, which is a...

11:42 — this decomposition of M_{tilde} . So we have an initial value M_0 . Then we have the non anticipating variable A_1 that depends only on information at time 0 times the increment of the Martingale from time 0 to time 1. And we add successive terms. So these A 's basically are random variables at the beginning of time. But after observing, say, j time periods, the A_j , let's see, will be-- or j minus 1 time periods, the A_j will be defined and non-random going from...

23:25 — So we have this problem, and we want to know what's the likelihood of basically player one losing. And the S_n is a martingale. And τ , the time at which the sum S_n equals plus A or minus B is a stopping time. And so this is a martingale by the stopping time theorem. And with such a martingale we have that the expected value of S at the stopped time is equal to the expected value initially. And that is equal to 0. Starting...

33:24 — And if we solve for the λ value, that makes ϕ of λ equal to 1. Then for that fixed value of λ , this M_n is going to be a martingale. It's actually a martingale for any λ . But in addition, that special case where ϕ of λ equals 1. And so if we graph this moment generating function as a function of λ , we can see that it's equal to 1 when e to the λ is equal to q over p . So M_n ...

44:56 — And to define the probability distribution of the entire process x_n using this stationary transition probability matrix and initial probabilities, we can compute any probabilities by computing the collection of all such probabilities where we have n steps

of the process, on n time points of the process, and states i_0 through i_n there. So if we can calculate these probabilities, we will have specified the probability model. Now, in calculating the joint probability of the first n values states being i_0 ...

56:54 — and a down day followed by a down day. And look at trying to model the state changes for a two day state with a Markov chain. And in R, there's a nice `r` package that will do this for us. And this was distributed in the Canvas site. And so you should be able to replicate this example with Apple or choose any other stock you might be interested in. And so importantly, this uses the library `quantmod` here. Sort of hard to see....

67:41 — the notation and setup for multiple linear regression. If you've taken any statistics class prior to this, which most of you have, you're very familiar with simple linear regression where you have one dependent variable and one predictor or explanatory variable. We want to consider the case of multiple linear regression, where we have a single dependent variable and multiple predictor or explanatory variables given by x 's, and we want to describe the relationship between the dependent and independent variables. Now, different regression problems can...

78:57 — So that for me is a guiding question throughout these different checks. So with that, we'll finish today. And next time, we'll finish these notes going through ordinary least squares estimation. And what's useful about this development of least squares estimation is that it does so with linear algebra. And what we'll see is that with linear algebra, we get very straightforward formulas for least squares estimates, as well as interesting analytical properties of those least squares estimates. So we'll cover that next time.

4. Câu/ý tiêu biểu từ transcript

"The model assumptions we apply include what are called Gauss-Markov assumptions, which is where we assume that the errors in the model have mean 0 and constant variance."

"Now, this second example of a martingale where we have a random walk S_n but then we square it and subtract n times the sigma squared, the variance of each of the steps, this is also a martingale."

"Now, this issue of completely specifying a stochastic process is an important one, in that what we want with a stochastic process is to have a complete probability model of that process."

"And so those conditional probabilities, together with the marginal probability, specify the distribution of the process over state 0 and 1."

"And so if we take the expectation of those from time 0, then by conditional expectation properties, the A 's are fixed given the information up to time $j - 1$, and then the variation of the martingale process from time $j - 1$ to j has expectation 0."

"With the example here of a simple random walk with equal probability of going up or down one step, one can graph the standard deviations up from the average 0 for the processes."

"We think of this set or event being the underlying probability distribution states where the tau variable is less than or equal to n ."

"We can generalize the Gauss-Markov to assume non-constant variances of the errors and possibly correlations of errors over time."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 7: Linear Rates, Products, and Models

Source transcript: mit_18_642_transcripts\08_RvXwSoGDYvg.json

1. Chủ đề nhận diện trong video

rates/bonds

machine learning

black-scholes/options

event markets

portfolio/risk

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Trọng tâm liên quan lãi suất/trái phiếu: chiết khấu, yield, đường cong lãi suất, duration hoặc sản phẩm rates. Đây là nền cho định giá dòng tiền và hiểu môi trường lãi suất. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:16 — SPEAKER: So it's my pleasure to introduce Andrew Gunstensen, who I had the honor and pleasure to work with for a long while at Morgan Stanley. And Andrew is actually MIT alum. He got his PhD in Geophysics from MIT. And after that, he worked in a variety of roles in financial-- senior quantitative roles in financial industry. And now Andrew is head of quantitative strategies at Mizuho in New York. So Andrew, welcome, and please take it away. ANDREW GUNSTENSEN: OK. Thanks, everyone. So...

09:15 — I'm not picking at you. You may have a terrible credit rating, so you have to pay 8% to borrow money. You may have a medium one, so you're borrowing at 6%. And it all depends on the particular issues of the issuer. So this makes it interesting because, at this point, a contract for you guys to pay me \$1,000 bucks in a year has different prices depending on who you are. So there's no single price here makes it interesting. A lot of...

17:32 — You want to pay every three months or six months or annually, for example. That just makes it just easier mechanically to deal with. You can compound it into term rates. So we'll talk about that in a slide or two. Another thing, if you do compound it into term rates, so you do like the average SOFR over a three-month period and then pay that rate. You don't know the rate until the end of the period because you haven't finished doing all your...

25:53 — is an interesting feature, which means it's an entirely one-way market. Because banks issue loans. They have to hedge it. It's all in the same direction. So there's a very substantial basis between regular vanilla SOFR swaps and the term SOFR swaps, something around four or five basis points, which, again, in a market context, where that's 20 or 30 times a bid offer, it's way outside the bid offer range. There's another older version of SOFR called FedFunds, which is really an average of...

34:20 — There's all sorts of crazy stuff. And imagination is the limit here, as long as you convince somebody else to do the other side of it. They're a tremendously useful instrument. They're great if I need to hedge something or change my risk profile, it's a very efficient way to do it. These are all derivative products, means they generally don't

use balance sheet. They don't take a lot of upfront capital. They're cheap to fund. So they're quite popular for those reasons. And I...

42:40 — months, and another guy who can then fund for 20 years because he's a high credit rating, we can both borrow in our respective countries and then do a swap. So I now have fixed rate swaps, and he has floating rate swaps. And depending how they want to manage their treasuries, that may be a great trade for all sides. So yeah. Again, swaps are very flexible and let you move the risk around in quite a flexible way and quite a customizable way....

50:43 — So there's another curve model that I think is probably pretty much-- pretty widely used at this point in the industry, which is I'm going to build my knot points based on the central bank meeting dates, because those are the only dates that rates are going to change, in general, or going to change materially. And I'll assume it's flat in between. So that's one. Again, I think that started six or seven years ago. And it's pretty-- far as I know, almost everybody...

58:45 — Did I do unwinds? Did things get assigned to me? Do I have resets? Did I exercise options? All of that will have an impact. There's the passage of time. That's the complicated one. So I went home yesterday, and I came in today. Market hasn't moved. But what does market not move actually mean? It's a very subtle effect because interest rate curves are a function of time versus this. So when I move this date to that date, what do I hold constant,...

66:51 — You can't do too many of those, or you're not trading anymore. So you have to correctly account for all of these extra costs, and that's quite non-trivial, just both mechanically in their system side, as well as conceptually. And a lot of these things are portfolio effects. So how do I then allocate that to individual trades is not necessarily super obvious. Good. Coming to the end at about the right time, quick blurb about electronic trading, which is-- if any of you guys...

74:53 — the deal had gone away. So that was a bit annoying. I think I'll end there and just open it up for any more questions. I think we've got a couple of minutes still, so I'm happy to yack about anything, as I said. So otherwise, I think we'll call it a day there. Skip through some bonds. AUDIENCE: I guess like one of the main things that people have been talking about is the interest rates in the recent quarter. What is your opinion...

4. Câu/ý tiêu biểu từ transcript

“These are not huge, multi-factor, stochastic calculus models you have to do and term structure models and things.”

“So there's a very substantial basis between regular vanilla SOFR swaps and the term SOFR swaps, something around four or five basis points, which, again, in a market context, where that's 20 or 30 times a bid offer, it's way outside the bid offer range.”

“So let's talk now about the core valuation model really in interest rate swaps, which is the yield curve.”

“And again, I said-- they mentioned they're-- basically, they're complex and inputs to the more complicated model.”

“So given all that sums to zero, I can work out what the implied forward rates from this discounting curve are.”

“So again, you can do a parallel gamma or factor gamma, a number of other approaches, or just use scenario analysis.”

“So again, if I'm doing a 30-year swap, it's now 3 and 1/2 basis points between them.”

“So I'm going to use just simple PV01 is my main risk measure, which is just a sensitivity to one basis-point increase.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.

- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 8: Regression Analysis (cont.)

Source transcript: mit_18_642_transcripts\09_cMF_c2WNPYU.json

1. Chủ đề nhận diện trong video

linear algebra

probability

regression

portfolio/risk

machine learning

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần đại số tuyến tính giúp biểu diễn tài sản/danh mục/dữ liệu bằng vector và ma trận, là nền cho PCA, covariance và tối ưu danh mục. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần hồi quy dùng để kiểm định quan hệ giữa biến giải thích và kết quả, nhưng phải cảnh giác overfit và in-sample bias. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:11 — PETER KEMPTHORNE: All right. Well, let's continue with our discussion of linear regression modeling. And as we introduced last time, there are ways one can formalize fitting a model, which is to propose a model, specify criteria for judging different fits or estimators of model parameters, then find the best estimators, and then check our assumptions underlying the specification of the criterion. And possibly, if necessary, modify the model because the assumptions that we made aren't satisfied. So we either want to add additional assumptions...

12:22 — verify that it's a projection. And a very important property of this residual vector is that it is orthogonal to the fitted values. So in the normal equations, we have this equation must equal zero. This term here is the residual vector. And if we multiply that n vector by the transpose of x , and it's equal to 0, then we must have that the error vector, ϵ , is orthogonal to x . Now let's see. I'm not sure if it's on the next slide...

24:32 — And this is going to be the expected value of E to the τ transpose β . So this is the definition of the moment-generating function. And if we define the matrix A to be basically the premultiple factor of A that gives us β , then our moment-generating function for β can be written as this expression, just plugging in AY for β . But then we can express τ transpose A equal to t transposed. So this moment-generating function of the least...

36:45 — And we get n minus p sigma squared for the expected value of sigma squared. And in regression models with normal linear models, we use this relationship to estimate sigma squared. Dividing both sides by n minus p , we get an unbiased estimate of the error variance. Now an additional property that's from part C of this theorem is that the error vector, ϵ , and the regression parameter, β , those happen to be independent of each other. And the independence follows by looking...

48:38 — every pair of variables in the data set are plotted in a scatter plot with the other variables. And so along the diagonal, we have the names of the variables, and then we have different relationships of those. Now if Y is the dependent variable of interest, if

we look of across this row, we're looking at the scatter plot of different independent variables with that. And it looks like the lca vol is probably a very strong relationship. Some of the variables, in fact,...

61:07 — And so it basically tells us how predictable our dependent variable is, given the independent variables. Now in terms of evaluating the assumptions of our regression model, there are regression diagnostics that we can apply to fitted models and there are a number of different important measures that we can apply. And in the stats package, there's this influence measures general function, which gives a table of different statistics. There's this R student for computing studentized residuals. If we have ϵ hat, it is distributed...

72:46 — it's quite general. But think about this. If we have the constants c_1 through c_p correspond to specific values of the explanatory three variables corresponding to a different case, then we can be estimating the true mean value for a given case of the x variables. If we have two x values, x vectors for different cases, and we're interested in what's the difference in mean values for those two cases, then we can have these c 's represent the difference in the x 's. And we're...

4. Câu/ý tiêu biểu từ transcript

“So if we have our regression model for our y vector and X matrix, we set that up as the conditional expectation is $X\beta$ and the covariance matrix is equal to σ^2 times the identity.”

“Here is what's called a quantile plot of the residuals that-- where the quantile plot will follow a straight line if the data are consistent with the normal distribution model and the t -distribution for normalizing the residuals by estimates of the variance.”

“So what we can do is consider generalizing the Gauss-Markov assumptions to, again, have 0 mean errors, but have the covariance matrix of the errors be given by σ^2 times a capital sigma matrix.”

“So we're basically looking at the distance of $\hat{\beta}_i$ from the true β and normalizing that by the covariance matrix.”

“And the error vector, ϵ_1 to ϵ_n being iid normal, this, in fact, is consistent, or is equivalent to a multivariate normal distribution of dimension n with a 0 vector for the mean and a covariance matrix that's diagonal with constant variance.”

“And so because of that, when we look at, for example, estimating the error variance σ^2 , well, if we take the sum of the squared residuals, that's the trace of the covariance matrix of the residual vector.”

“And it will turn out that this conditional distribution of y given x and β will have a multinomial distribution with a mean vector μ and a covariance matrix σ .”

“And if we simplify it, we basically have a mean vector, β , for the least squares estimate, and a covariance matrix for the least squares estimate that's given by $\sigma^2 x^T x^{-1}$.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 9: Principal Component Analysis in Finance

Source transcript: mit_18_642_transcripts\10_CechARGinR4.json

1. Chủ đề nhận diện trong video

[pca](#)[machine learning](#)[portfolio/risk](#)[linear algebra](#)[rates/bonds](#)

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Trọng tâm liên quan lãi suất/trái phiếu: chiết khấu, yield, đường cong lãi suất, duration hoặc sản phẩm rates. Đây là nền cho định giá dòng tiền và hiểu môi trường lãi suất. Phần đại số tuyến tính giúp biểu diễn tài sản/danh mục/dữ liệu bằng vector và ma trận, là nền cho PCA, covariance và tối ưu danh mục. PCA giúp tìm factor chính và giảm trùng lặp dữ liệu, rất hữu ích khi nhiều indicator kể cùng một câu chuyện. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:17 — PROFESSOR: All right, I want to welcome Stefan Andreev as guest lecture today. He has been a significant part of this course over the past, I think, 11 years. Is that right, Stefan? STEFAN ANDREEV: A long period. And I can't count that far. PROFESSOR: Partly because his lectures are the most popular in the course, and we really are glad to have him come back again. His background is very interesting. He has a PhD in Chemical Physics from Harvard. And he has worked...

11:51 — In the real world, when you do-- it could be in science or in finance. You're not going to have perfect data. Everything is very noisy. And noise is a huge problem. And robustness to data noise is very important. So this illustrates how robust things are. So if I take one point and I just move it around, you can see that the red line, which is the direction of my PC1, doesn't-- it's quite robust. It's quite robust to noise, to one particular....

22:55 — What would be a concrete example of where it would be better to not normalize it? STEFAN ANDREEV: Well, here's an example. So actually-- yes, I will start with this. So this is an example of-- it's a little bit of a toy example. And I'll talk about a real example too. So in this toy example, there's the blue and the red data set. And the red one looks like it's one-dimensional. And the blue looks like it's just a blob. There's no direct...

34:38 — Second, yes, intraday observations are-- they represent additional data. However, the quality of the imagery that data, meaning how much noise there is, can vary often, in practice, be much higher for intraday data than it is for end of day data. That may not always be true. There are situations where that's not true. And there's entire fields in finance that deal with high-frequency trading in those fields end of day data is just like their whole trading, buying, holding, and selling happens 10...

45:49 — It's going to tell us something about the relative volatilities of points. So when we do that and then we look at-- what I'm plotting here on the top are the eigenvectors,

so the principal component factors, the shapes of the principal component factors on various days. So this plot can be confusing. And I want to make sure that we're all on the same page what it is. This is the yields. And we take these yields. We look at the changes of yields....

56:44 — is PC2 and above, what can I say about those returns? How can I model those? And the key thing-- you can construct these things. You can say, oh, I want to model these things. But the key is, once you have a good model, how do you trade them? So being able to construct-- to understand how PC loadings translate to portfolios is very important because it tells you how I can use this methodology to construct specific portfolios that are uncorrelated to each...

67:15 — was probably twice as high. So people expected the standard deviation of daily move that was priced in was probably 10 to 12 basis points per day. Does that mean that all this is irrelevant? No, it's not. It just means that when you're thinking about the risk you're taking, you have to understand that if you take a certain amount of-- for the same number of bonds or the same number of the same investment that you hold, if you hold it on that...

77:30 — basically, once you hedge out, the curvature and the levels, what remains is-- once you hedge out the slope and levels, what remains is the curvature. And the question is, at that point, how does the curve move? And I can give you two examples. In dollars, you can see that most of the time, the bottom of this graph is at around 10-year maturity. So that means that once you hedge out slope and curvature, 10-year bonds are going to be anticorrelated with 2-year...

4. Câu/ý tiêu biểu từ transcript

"There is also a more efficient method called SVD, Singular Value Decomposition, where you can just compute your projections without calculating the covariance matrix."

"Well, there is-- you could do-- you could take your data, compute covariance matrix, feed it into an eigenvalue calculator, which is basically-- in my work, our data is not really-- the number of dimensions is not actually that high."

"Then you're going to compute your covariance matrix and calculate the eigenvalues."

"So we did PCA on covariance matrix, and the reason is that the units, the yield changes are comparable units."

"The covariance matrices can be written as the-- in this form, where lambda is the eigenvalue matrix."

"So when we do that and then we look at-- what I'm plotting here on the top are the eigenvectors, so the principal component factors, the shapes of the principal component factors on various days."

"So that adds up to 100% when you're-- but that's very useful because if you remember how I was saying want to look at the eigenvalues to understand how important each direction is, variance explained tells you that as a number."

"I like to compute the covariance matrix, because we generally need to compute the covariance matrix for other things in any case."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 10: Counterparty Risk Optimization

Source transcript: mit_18_642_transcripts\11_VbtXo62ROC4.json

1. Chủ đề nhận diện trong video

portfolio/risk

machine learning

black-scholes/options

probability

linear algebra

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần đại số tuyến tính giúp biểu diễn tài sản/danh mục/dữ liệu bằng vector và ma trận, là nền cho PCA, covariance và tối ưu danh mục. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:11 — PROFESSOR: All right. Well, today's guest lecturer is James Shepherd with Quantile-- JAMES SHEPHERD: Quantile, yeah. PROFESSOR: --which is a wholly owned subsidiary of LSEG. And your background in quant finance is quite long. And so I'll let you share some of the highlights. JAMES SHEPHERD: OK. PROFESSOR: Thank you very much, James. JAMES SHEPHERD: Thank you very much. Do I need that. PROFESSOR: No. JAMES SHEPHERD: OK. Thank you very much. Thank you for having me. This supposed to finish about-- so a quarter...

10:23 — on the two portfolios I had in the first place, which seems kind of sensible. If I put some stuff here and some stuff here and put it together, it seems unlikely that I've manufactured some risk out of doing this. But now conversely, it seems highly plausible that there might be some diversification of putting these two things together. And so my overall risk might well be smaller. It seems a very plausible and reasonable property to ask for from a risk measure. So...

20:22 — In reality, it seems that the losses-- the changes that you'd have on a typical portfolio or a typical market data are not independent. If you lose a lot of money one day, it's a reasonable chance you might lose some money on the-- or the market might go down on the second day. So there's a thing called autocorrelation, which is basically the correlation of a time series with itself shifted by one day. And so you can-- which is what this thing here....

30:49 — And what is the probability that I'm going to see 12 or more-- so there's 12 violations, by the way. So if I take this 225 and say, how many times did it actually-- according to that period, how many times was it actually-- the loss was bigger than 225, or there were 12 scenarios bigger than 225? So what is the probability that I would see 12 or more days out of a sample of 500 that exceeded the VaR? So that comes from...

41:19 — in the beginning of December. And that'll be the first one that's calibrated without any of the COVID or global financial crisis data in there. And what's going to happen, what people expect to happen, is that the initial margin calculated by this SIMM

is going to massively reduce by somewhere in the region of 5% to 20%. And what that will do is release something in the region of \$40 billion of margin back out into the wild. Because today, people are paying initial...

50:35 — What we're in the business here is we're solving minimization problems or operational research type problems. They're of the form minimize F subject to some other stuff. And as I mentioned earlier, solving complex problems is way, way, way easier than solving general, nonlinear problems. And convexity is basically the same thing as subadditivity. That's the definition of convexity. And that should remind you of the definition we saw earlier of subadditivity. So we saw earlier minimizing expected shortfall. That's basically a piecewise linear problem....

60:14 — you can solve it in 11 seconds and with very, very few iterations, which is basically a testament to the amazing power of Newton-Raphson quadratic convergence. So what happens when you try and solve it? This is how you have the initial risk, and this is with the optimized risk on the right-hand side. As you might expect, it tries to form-- it tries to basically get it, but I don't have some positive numbers and some negative numbers. It tries to nestle the risk...

70:04 — put more and more banks or parties into the system, the savings should go up. That's kind of a reasonable thing. But more than that, the saving divided by the initial, that also goes up. So you get not just the total saving in the system goes up, but the total-- the efficiency increases, the more people you add into the system. I wouldn't put too much faith into whether it's really linear or whether it's a kind of not linear thing in there. But...

80:10 — That's the definition of VaR. So if I'm trying to come up with a model where I expect something to happen 1% of the time, or I expect the VaR to be exceeded 1% of the time, then I do 500 draws of that thing, and I say, well, how many times was it actually exceeded? And then the probability of that being 12 is 5%. AUDIENCE: So you are being conservative. Like this-- JAMES SHEPHERD: Yes. AUDIENCE: This VaR is conservative. Therefore, it's OK....

4. Câu/ý tiêu biểu từ transcript

"So all you need to do for the model-building approach is figure out what the variance-covariance matrix is between these different elements, which I computed here."

"So there's a variance-covariance matrix between the-- right-- that you get."

"If you were to make the assumption that the change in the portfolio-- so this delta P is a normal distribution, then VaR and expected shortfall would have these two formats here."

"So the historical VaR-- those equivalent numbers were 320 and 377, basically showing that these normal distributions, it's clearly not a normal distribution."

"So once you've got a type of risk, once you've got some risk, generally speaking, you want to mitigate it, i.e., reduce either the probability of it happening, or reduce the impact of that risk happening, not just for counterparty risk, for all kinds of risk."

"But they still require you to back test VaR So what you generally do is you build a model, which contains-- which will predict both VaR and expected shortfall."

"And the general rule of thumb is that if the probability is 5% or more of getting that degree of-- that number of violations of the VaR, you say that the model is OK."

"And if the probability is less than 5%, you'd probably reject the model and say the VaR is not a good model."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.

- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 11: Regression Analysis (cont.)

Source transcript: mit_18_642_transcripts\12_RruxdEjlvv0.json

1. Chủ đề nhận diện trong video

regression

machine learning

portfolio/risk

probability

black-scholes/options

2. Giảng lại đầy đủ ý chính

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Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:13 — PETER KEMPTHORNE: All right. We'll get started. What I wanted to do today was to make sure we thoroughly cover the theory and application of regression modeling. And so to begin, there's a recap, which is the theory for, if we have linear regression, and we have, in particular, a normal linear regression model, then we have these properties of beta hat being distributed as a multinormal random variable with mean the true regression parameter and covariance matrix given by the X transpose X inverse...

11:52 — Then we compute the density of our data Y , given the explanatory variables in X , and the beta, regression parameter beta. That should be a beta there. And we simply identify those parameter values that maximize this probability. So we're looking at the parameter values that make the data we observed the most likely. And intuitively, that's maybe a reasonable thing. If you take Mathematical Statistics next term, you'll learn that that, in fact, is an optimal estimator, in terms of having smallest variance in...

23:30 — And so in order for this to be a reasonable thing to do, we actually need to-- or, we don't need to, but it's useful, to consider standardizing our independent variables x , making them have mean 0, and also centering y , and then consider the regression where we rescale the columns of the predictor matrix as centered with the inverse of the covariance matrix between those. So in this formulation, this is the statement of how the ridge regression parameter estimate is specified. Now, the...

36:02 — Then we basically have diamonds where, along each of these diamond edges, the penalty is the same magnitude. And so with lasso regression, we basically consider finding an estimate that minimizes the least squares but also is constrained to have sort of sum of absolute magnitudes that's smaller. And so this kind of penalty ends up actually leading to shrinkage, which tends to concentrate at the vertices of this penalty function. So what we end up getting, often, is a lasso estimate that has some...

48:08 — include, really, necessarily, the important regressors in the model. So we can have statistically significant high-order principal component variables that don't have much variability in them, but they actually are explaining significant factors affecting the

response variable. So anyway, this note goes through and additionally considers ridge regression and lasso regression. And so with the ridge regression results, the display of regression parameters is traced out as we consider different constraints on the L1-- or on the-- well, on the L2 norm. And if we...

59:45 — What could be useful is testing whether our beta coefficient is consistent with sort of the average market risk of beta equal to 1, or not. But here's basically a plot of the linear regression model for GE, and a discussion just of getting comfortable with R-squared values and calibrating our views of scatter plots to those values. And then there's residuals analysis, that is very important. As I introduced at the beginning of our regression study, we make assumptions about our model. And we...

70:37 — just to see how they play out. But let's see. What's interesting, I guess, is, with this change between two periods, if we consider sort of an original regression model, y equals X beta plus epsilon for a very long period, we might consider splitting it up into the first part, period A, and the second part, period B, and have two equations for the respective sets of data. And in thinking about the two periods, we basically can test whether the regression parameters for...

81:58 — a huge amount of returns. But what's interesting to see, though, is that this relatively simple model, the capital asset pricing model, appears to be consistent with the stocks over a long period of time. That being said, the more complex models are warranted. And the data collection phase of this, our project, included importing Fama-French factors that are extensions of sort of the market risk that can be used in these models, so.

4. Câu/ý tiêu biểu từ transcript

"But with the normal-- or with our testing of individual coefficients, we basically use the fact that our least squares estimator is multivariate normal with mean equal to the true beta vector and covariance matrix, given here."

"And we also have that epsilon hat is a multinormal distribution-- it's actually not of full rank, but it's in n dimensions-- with mean 0, a 0 vector, and the covariance matrix is sigma squared times the identity minus the hat matrix."

"So this ridge regression is interpretable as putting a prior distribution on the betas that have this form, mean 0, and covariance proportional to the identity matrix."

"So with our data predictor matrix X , we consider the sample covariance matrix of those row vectors."

"And with these eigenvectors, which are normalized to have length 1, we simply multiply our matrix X by the j -th eigenvector, and we get our j -th principal component variable."

"And for maximum likelihood estimation, we use the probability model of the regression-- which, in this case, is normal linear regression models, so if we have normal epsilon vector."

"So we can have statistically significant high-order principal component variables that don't have much variability in them, but they actually are explaining significant factors affecting the response variable."

"And with principal components regression, one has principal component variables which are orthogonal to each other, and so the computation of their coefficients is very simple."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 13: Portfolio Management

Source transcript: mit_18_642_transcripts\13_o7OnkMdmjLg.json

1. Chủ đề nhận diện trong video

portfolio/risk

machine learning

black-scholes/options

probability

volatility

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Volatility modeling cho thấy biến động thay đổi theo thời gian và ảnh hưởng trực tiếp đến risk, sizing và option/CW. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:16 — JAKE XIA: So today, we'll be talking about portfolio management as one of the application lectures. So I will share some of my research work from a practitioner's point of view, rather than from pure theoretical or math point of view. So as we mentioned at the beginning of the term, we've been teaching this class since 2012. And in 2013, this class was recorded for MIT OpenCourseWare. The link, I put it here. It's the lecture from 2013. Some of you may have watched...

08:25 — how to compare different investments. To understand the relationship between different investments is to understand the return and the risk profiles of all the investments. So what is risk? So in this chart we are talking about here, I'm using volatility for the time being, which represents uncertainty of the outcome. Later on, I will come back and tell you volatility is not really the best measure for risk. But let's save that for later. But before we talk about the investment choices, let me...

17:19 — If there are competing offers, how much more you can offer to the professor to get them to come to your university. So there's definitely absolute return angle for the requirement of the portfolio. And there's also a lot of people nowadays talk about beating a benchmark, beating an index. Many of you have chosen S&P as an index. But if you can produce more return on top of S&P return, that's called the alpha, relative return. That's also often being used as a measurement...

26:02 — look at everything, and also specialists in certain domains, such as biotech, AI, and so on. We focus on both absolute return and also relative return to the benchmark. So we are mostly active managers. So we have some passive beta index investments. But it's a minority of the investment. It's also not always permanent. So whenever we can find an active strategy or manager to beat a benchmark or index, we will always go for the active investment. Yeah, so as you can see,...

35:33 — But it's a straight line, in this case. You can see the slope of that line is determined by the difference of R_2 minus R_1 and divided by σ^2 . So by the way, this is so-called capital allocation line in the portfolio theory. So we can look at, let's say, a

constructed portfolio. You already have a σ_p , which has many assets in it, and also R_f . By adding one riskless asset, or cash, which returns R_f . R_f is basically like a risk...

44:49 — So people say there's a free lunch. The only free lunch is diversification. But you have to rebalance to eat it. So that's on that. So then I'm going to shift the gear to talk about the limitations of so-called modern portfolio theory, in short MPT. I mentioned this was mostly developed in the 1950s. And Harry Markowitz pioneered in this area. He actually got a Nobel Prize in the, I think, early '90s for his work. So all I have just described to you...

53:47 — may be good. But the denominator will normalize it to make it basically comparable to other investments. So the optimization problem-- yes, please. STUDENT: So in practice, what do you use to estimate the expected gain? JAKE XIA: Yeah, bless you. STUDENT: estimate the distribution of the pricing? JAKE XIA: Yes. But no more difficult than getting the return and standard deviation. Yeah, so it's the same amount of work. It's just that you're separating the numbers. You're calculating half distribution instead of the whole...

64:00 — get into the reactive state. So the whole crowd becomes synchronized. The observation basically shoots through the roof. So it becomes a very unstable system. So that's the bubble. At that point, when the system becomes unstable, your bubble will burst. So that's the part of the crowding behavior. So the purpose of showing that is, when we look at portfolio management, this type of bubble forming and bursting process is a very important aspect of our understanding beyond the fundamental analysis. This is all...

73:09 — But I would say, nowadays, it's definitely the governments. So you need to understand the key drivers of your portfolio. So I want to stop the lecture here. I want to take some questions. Then I will save a couple of minutes to talk about the investment game. But before I get there, let's open up for questions. I packed in a lot of content for today. I know. The idea is not necessary to ask you to follow in details because some of the...

81:01 — Maybe there's something beyond what we talk in this class that could be useful to any one of you. So with that, thank you very much.

4. Câu/ý tiêu biểu từ transcript

"So I will talk about my own research work, in terms of how to improve the portfolio construction theory-- for example, how to improve the risk measurement using volatility and Sharpe ratio and how to model crowding behavior."

"That's very much similar to the risk parity portfolio, which they try to make equal risk contribution coming from the bonds and equity by borrowing more to invest in the bonds because bonds typically have lower volatility than equity."

"So when we add the risk free asset on this R_f point, we can basically combine that risk free asset with the existing portfolio to further improve the risk-- the efficient frontier to move up to that straight line."

"So in mathematical terms, the probability of b times x -- x is the random variable."

"So for a random variable x , the normal distribution we're all familiar with is the e to the x minus the mean divided by 2 sigma squared."

"So basically, the random variable x versus the rank of X , which is more displayed in this chart."

"Or do you think you want more lectures on the research process, how to do data collection, signal generation, backtesting, and all the things and the portfolio optimization, as we talk about today?"

"And the variance-- or in this here, it's the σ_p squared, the volatility squared, can be expressed in terms of the weights and the volatility of each of the two investments, as well as the correlation, ρ , between the two investments."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.

- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 14: Stochastic Processes II

Source transcript: mit_18_642_transcripts\14_VM29Jy1sio.json

1. Chủ đề nhận diện trong video

probability

stochastic processes

black-scholes/options

stochastic calculus

portfolio/risk

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần quá trình ngẫu nhiên xem giá là đường đi theo thời gian, nên drawdown, hitting time và thứ tự dữ liệu rất quan trọng. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nổi sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:12 — PETER KEMPTHORNE: Today, we're going to talk about stochastic processes, the second set of lectures. And we're going to begin with a discussion of Brownian motion. Brownian motion is a process that I'm sure most of you have heard of. And it was named after Robert Brown, an English botanist. And this is an interesting case where a scientist observes some dynamics in nature and then is concerned about what mathematical model might describe that behavior. Turns out that it was, I guess, Albert Einstein...

10:19 — And the cumulative distribution function is simply the integral of the density from minus infinity up to c for ϕ of c . And this notation of small case ϕ and uppercase Φ are standard for referring to Gaussian or normal densities in cdf's. Well, if we have a normal with non-zero mean μ and variance τ^2 , or standard deviation τ , then we have the density of this normal Y is a simple function of the density of the standard normal. We simply subtract the...

21:52 — X's and Gaussian X's, which of course will be Gaussian. And so here is a cumulation or a random walk with 10 steps. The top panel corresponds to unit plus 1 or minus 1 steps, and the bottom panel corresponds to Gaussian steps. And if we increase the number of steps and normalize them by dividing by root t , we end up getting these different paths tracing out. And so what's important in this display is that, as we let n get arbitrarily large, the...

33:48 — for that time, which is relatively short. But if we're looking at hitting times for higher levels, like two or three, then that hitting time distribution spreads out. So now, let's see. We can talk about some extensions of Brownian motion. One is just reflected Brownian motion around 0. So if we have the process bouncing off 0 level, we can think of that process being a positive stochastic process. And it's fairly easy to show that the expected value of this reflected Brownian motion...

46:48 — ends up being equal to-- well, it's exactly equal to the variance of that increment plus the square of the mean. But the square of the mean is a term that is small order of Δt . So our expected squared increments will be to order Δt equal to

$\sigma^2 \Delta t$ and then a small order of Δt term after that. Also, if we consider powers of Δt , so our increments to a power, then when our power c is...

59:13 — if μ is equal to 0, and it's equal to this constant, A times e to the minus $2\mu x$ over σ^2 . All right. Well, this general solution has to satisfy the initial conditions. u of a is going to equal 0, and u of b is going to equal 1. So if we start at a , where the-- basically, we stop at a , if we start at a . So u of 0 is going to equal 0. And if we start at...

70:55 — basically, for prices of assets and equity markets, this corresponds to the log return from time 0 to t . And that log return from 0 to t might follow a Brownian motion process with drift, positive return. And so we basically can have this process evolving. If we were to calculate the quadratic variation of this process, then if it followed-- if it followed a Brownian motion process, it would tend to grow according to a straight line, with slope equal to σ^2 . And...

4. Câu/ý tiêu biểu từ transcript

“So our process X , Brownian with drift, will have non-zero mean increments and increments variances that are proportional to the change in time.”

“And so when we look at the covariance of the process at times s and t , it's equal to the expected value of the product.”

“And so it turns out that Brownian motion can be defined as the limiting stochastic process corresponding to a random walk.”

“So with this normalized random-walk process, it turns out that this limiting distribution is-- or this limiting process is invariant to the distribution of the X 's.”

“So these are the normalized random walks, which are effectively Brownian-- or whose limits are Brownian motion.”

“So let's let X of t be Brownian motion with drift, μ and variance σ^2 .”

“So if we have random walks with independent steps that have mean 0, standard deviation 1, then regardless of what that distribution is, the limiting normalized random walk will be the diffusion process.”

“And the normal distribution for the magnitude of that increment will be a normal random variable with mean 0 and a variance that is proportional to the length of the interval.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 12: Time Series Analysis

Source transcript: mit_18_642_transcripts\15_qlytPllimpQ.json

1. Chủ đề nhận diện trong video

linear algebra

machine learning

portfolio/risk

probability

black-scholes/options

2. Giảng lại đầy đủ ý chính

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Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:11 — PETER KEMPTHORNE: So today we're going to cover an introduction to time series analysis. And time series analysis is really a very powerful methodology in statistics, and especially powerful when it comes to modeling financial markets. So I hope you'll find this introduction engaging and motivating. So let's talk about time series. We basically have notation indicating a stochastic process where X corresponds to the value of the process at time T , and the time periods T can be discrete. This is most commonly what...

11:15 — And so what I'd like to do next is to look at some time series plots of different financial time series, and see whether this transformation to log returns seems to make the series appear stationary. So here's the S&P 500 index. These graphs here actually update the graph from before to the end of September. And if we change initially the absolute price to the log of the price, you'll notice that an exponential growth tends to transform to a linear growth on the...

21:34 — which isn't too different in its appearance. But then if we look at the changes in log, the increments in log, which are log returns-- in mathematical definition, log returns are not really the right terminology for the percentage changes in yield. But we'll basically be using the same terminology as before. One can see that we basically get a distribution of increments, steps in the random walk on the log scale that are unimodal and symmetric. And as we consider higher frequency-- here's weekly...

34:03 — to test if the model is good, basically. And a good model will be a model that has captured all of the time dependent information in it, and the predictions from these or forecasts from this model are ones we have confidence in. And so there's actually another assumption here, which is that-- let's see. If the epsilon t are normal or Gaussian, then this gives good predictions. And by predictions, I'm thinking of not point forecasts of a series, but a prediction interval and...

46:52 — learn about how to specify a moving average model to these series and get estimates of the size and also estimates of the white noise terms. And with this step, we

then can evaluate whether our residuals are orthogonal to lagged values for lags greater than p back. So these residuals should not have any dependence on lags of y that are further back than p . If those are not orthogonal, then we should increase p . And then we want to look at whether the...

58:35 — as being equivalent to applying the inverse operator to both sides. And so we get an inverse operator, which is a polynomial lags operating on X_t is equal to the η_t . So this corresponds to actually an autoregression model that has white noise terms. And so when this exists, this is the format of that. And what we'd like to do is show you how this works with some different examples. Right now it's perhaps a bit abstract, but when we look at the...

69:17 — So with an autocorrelation, the ϕ is the first order autocorrelation, and ϕ to the j -th is the j -th order autocorrelation. OK. Now, if ϕ is between minus 1 and 1-- for ϕ between 0 and 1, the process exhibits exponential mean reversion. And when ϕ is negative but no larger than negative 1 in magnitude, then it exhibits some oscillating exponential mean reversion. But when ϕ equals 1, then the Wold decomposition does not exist and the process is a simple random walk which...

80:20 — applied actually with different cases. OK. We'll finish there for today.

4. Câu/ý tiêu biểu từ transcript

“So if our model basically predicts up to this residual process, which is unpredictable, has mean zero, constant variance, is uncorrelated, there's no information left in the residual series.”

“With the covariance stationary process S_t , we have basically that μ is equal to the expected value of S_t .”

“With moving average processes, if we look at time increments j that are longer than the q order, then there's no overlap of white noise terms in X_t and X_t plus j , and so their covariance is going to be zero.”

“Well, if we have a time series X_t that we want to specify a covariance stationary time series model to it, well, the first thing we would do perhaps is de-trend the series X_t so that it has a constant mean level.”

“The constant expectation of the covariance stationary process is equal to C plus the sum of ϕ_j times μ , the expectation of each of the lags, plus 0.”

“So any covariance stationary process can be represented as this moving average of white noise process.”

“And with stationary processes that are covariance stationary, then it turns out that any covariance stationary process can be decomposed into a V_t process which is linearly deterministic.”

“And so we end up having these first and second moments and these covariances which are very easy to compute.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 18: Applying Data Science and Artificial Intelligence to Managing Biomedical Portfolios

Source transcript: mit_18_642_transcripts\16__e2nDnV7FQs.json

1. Chủ đề nhận diện trong video

machine learning

portfolio/risk

black-scholes/options

volatility

probability

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Volatility modeling cho thấy biến động thay đổi theo thời gian và ảnh hưởng trực tiếp đến risk, sizing và option/CW. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:16 — PROFESSOR: Today, we have a special guest lecturer, Professor Andrew Lo. A lot of you may have already known Professor Lo, but I still will make a quick introduction. So Professor Lo is professor at the MIT Sloan Business School for many years. He's also the director of the Laboratory for Financial Engineering and a principal investigator with MIT CSAIL, which is the Computer Science and Artificial Intelligence Lab. He also has many other active involvement with, say, just a couple of examples. Santa Fe...

11:05 — - Wow. Look at you trucking along. Come on. - She's actually walking, even though it's with a walker. She's actually able to walk. Now you can tell that there's still slow-- - She can't walk on her own yet. But this is only six months after the treatment. OK? Now this is just around the same time. She is now getting ready to actually stand on her own. And this uses an enormous amount of core muscles. You probably all know you have to...

20:52 — We'll see whether the budget discussions go well. As long as we agree to pay back our debts, yes, safe asset. But you know what? You put your money in it in 2008, you would have earned pretty much nothing. And actually, you would have earned less than nothing because of inflation if you look at the real return. All right. The red line that none of you picked, that is the US stock market. I bet you some of you may already have an...

31:05 — I don't have 30 billion, but you do in the aggregate. If I can make this investment attractive enough to get you to put some of your 401(k) money, or some of your summer internship money, into a fund like this, I can raise \$30 billion very, very quickly. Now the question is, can you really raise \$30 billion? And the answer, of course, is it depends. It depends on these calculations that I just did. It depends on the math underlying this portfolio. The...

40:49 — accomplishment that I've ever experienced was this moment here when I met that little girl. She actually lives in Boston in Jamaica Plain. And here she was. The parents found out about the business model that I had developed along with Neil, and asked to meet with me. And it was really kind of a scary and extremely emotional meeting. This is the Flour Bakery right across the Salt and Pepper Bridge, where Mass General is. That's where they were going for their care. And...

51:45 — that shows, yes, there is a path of creating a portfolio in order to fund fusion. Now a portfolio of what? So I have to answer that question. In order for me to answer a question about portfolio of what, I have to describe, what do I mean by saying, fusion? What do we mean by fusion? What I mean is to create a commercially-viable power plant, a reactor, that will generate enough heat to create a turbine that will spin and generate electrons to...

61:41 — for \$20 billion. Now that's not why I want to be Harvey Lodish, although that's not a bad reason, by the way. I'm an economist, after all. See, I want to be Harvey because of what happened in 2002. In that year, Harvey's daughter was pregnant with her second child, Harvey and his wife's second grandchild, a boy named Andrew. Great name, by the way. Andrew was diagnosed in utero with Gaucher syndrome. What are the chances of that? So I talked to Harvey about...

71:04 — looking to fund various different projects. So I would say that there are still lots and lots of smaller groups that are looking for funding that can't get it because they just don't know enough about the financing. So what I've described to you now, in just this one class, is probably more information than most startup groups have in terms of thinking about risk rewards, and trade offs like that. Yes. AUDIENCE: , you're also a professor here. So I was wondering and stuff...

4. Câu/ý tiêu biểu từ transcript

“And so when you calculate the variance of a sum of a portfolio of typical investments, you're getting not just the individual variances, but all the pairwise covariances.”

“If there aren't any pairwise covariances, that's where the risk reduction comes from.”

“If we don't have correlation, then can anybody tell me what the formula is for the variance of a sum of random variables?”

“You're born with a normal pair of eyes, but this mutation prevents your eyes from producing a very important protein.”

“So when you calculate the variance of the sum, it really is the sum of the variances.”

“So normal distribution, gamma distribution, chi squared.”

“But the key component is the fact that because these rare diseases are so different from each other, there's very little correlation in the successes and failures of different therapeutic programs.”

“And so if the variance is the sum of the individual variances, take the square root of that to get the standard deviation.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 19: Volatility Modeling

Source transcript: mit_18_642_transcripts\17_zapp8smQKhg.json

1. Chủ đề nhận diện trong video

volatility

machine learning

probability

portfolio/risk

black-scholes/options

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Volatility modeling cho thấy biến động thay đổi theo thời gian và ảnh hưởng trực tiếp đến risk, sizing và option/CW. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:10 — PETER KEMPTHORNE: Well, today we're going to talk about volatility modeling. And this topic in finance is one that has really interested me for decades. And there are a lot of subtle details involved with understanding volatility, how to measure it, how to forecast it. And we've had some talks already with the group projects about dealing with volatility estimation and measuring volatility, in particular implied volatility. Today we'll focus more on realized volatility and modeling that. And so to begin, here's just an overview...

11:03 — I think I have it listed here. So actually this RiskMetrics was created by JP Morgan back in the 1990s. Coincidentally, when I was at the Sloan School teaching statistics and financial modeling, I was part of the Financial Services Research Center that was sponsored by JP Morgan and other groups when this risk metrics document was created. And if we look at, let's see, there's a section here of risk metrics decay factors for volatility. Basically, for different countries, one can use the exponential...

22:02 — basically will have mean sigma squared and variance equal to 2 sigma to the fourth. Now, this corresponds to the theory of normal random variables for returns, which comes from the Brownian motion. And the expected squared value is-- well, actually, we can think of maybe the mean return being 0 or being subtracted from all the returns. But the variance of this sigma 0 hat squared is the variance of a chi squared random variable with 1 degree of freedom. And so in probability...

32:27 — Now, let's see. There's a case study of using R to calculate these different volatility estimates. And let's see here. So here's the case study, which goes through calculating different measures of volatility, using close to close Parkinson's all the way down to Yang and Zhang in section 1.6 and then comparing those for the S&P 500 Index. And we'll go through these in a moment. But following just looking at these individually, we consider time series modeling of these different volatility statistics. And so...

43:14 — estimate of what's going to happen is where things are now. So this naive model basically has an autocorrelation function, which has a very high-- or our data has a

high first order autocorrelation. And if we calculate first differences, which are the residuals from this model, then we get this residual diagnostic information, the time series of residuals, the autocorrelation function of the residuals, and the histogram of residuals. Now, would you say that these residuals are consistent with white noise that may be...

55:29 — we have the last term here. And what's generally important in fitting models is when you consider the last terms to enter into a model, such as in this case, the seasonal moving average term, its estimate is many multiples of its standard error. So that's a very important factor in the model. As an anecdote, let me just say-- let's see. If we take a series y_t and we look at a seasonal difference operator and a first order difference operator, this turns out...

66:31 — comparing it to the Laplace curve in blue. And an interesting question is, how do we judge whether one model or another is a good model? And a useful tool is to construct a q-q plot, which is a quantile-quantile plot. So we consider points where the vertical value corresponds to the magnitude of our data, and the horizontal coordinate corresponds to the theoretical value for the smallest from a sample of Laplace, the second smallest from a given Laplace distribution. And so we basically...

77:34 — in ARCH, the generalized ARCH model is capturing time varying volatility in an appealing way, whether that's really valid or not is of interest. But it certainly is an alternative to these arch models that is quite powerful. Well, here's just a graph of all three together. This note actually does a final section of looking at what happens if our error distribution is not Gaussian, but is a t distribution. There are packages, rugarch, which allow us to specify parameters and fits using t ...

4. Câu ý tiêu biểu từ transcript

“So one can find the long run variance of a GARCH model with this formula.”

“So by looking at essentially the percentage change on an infinitesimal level, we basically have a linear trend that's deterministic and then a random walk with Brownian motion steps on that scale.”

“One of them is a Poisson jump diffusion model, where the stochastic differential equation has the Brownian motion part as the first two terms, but then has a jump process as a last term where the jump process can be modeled as a Poisson process.”

“And there's a long run variance of the GARCH model that is given here.”

“With geometric Brownian motion, turning now to having a stochastic process model for the prices.”

“So if the error is ϵ_t are independent Gaussian distributions with different variances, then this likelihood function defines the joint density of our data given underlying parameters.”

“And with a GARCH 1, 1 model, there basically is a long run volatility σ^2 , which has this particular formula.”

“And so what's important is that when we apply this stochastic differential equation, there basically comes an extra term in the drift that's due to the squared variation in the process.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 21: Black-Scholes Formula, Risk Neutral Valuation

Source transcript: mit_18_642_transcripts\18_2UCHztIWuZg.json

1. Chủ đề nhận diện trong video

[black-scholes/options](#)
[portfolio/risk](#)
[probability](#)
[rates/bonds](#)
[machine learning](#)

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Trọng tâm liên quan lãi suất/trái phiếu: chiết khấu, yield, đường cong lãi suất, duration hoặc sản phẩm rates. Đây là nền cho định giá dòng tiền và hiểu môi trường lãi suất. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:17 — VASILY STRELA: So I'll talk mostly about risk-neutral pricing, and then Black-Scholes will be more of an illustration to the method-- to more general method, which I think is very important, and very important for pricing derivatives. And a few of you did look into Black-Scholes. So my hope is to give a little bit more depth and a little bit more perspective on what's the bigger picture of derivative pricing through risk-neutral valuation. But as promised, before, let's start from the interest rates from...

11:33 — So the payout is S minus K , where K is a strike. Agreed price. And for forwards, the agreement is that the strike is set in such a way that it costs you nothing to enter it today. You basically say, we agree with you today for a zero price that I'll buy the stock at the price K . So quite a useful diagram is the payout. That's-- sorry, that's the blue line, so it's-- in our case, it's S minus K . So that's forward....

23:20 — though, we can find the probability, which will give us our strike-- or which will give us this formula. So what will it be? Well, well K is equal F_0 . And so our P -- and I'll call it neutral risk-- neutral probability, S_1 plus 1 minus P_n S_2 . And we can find our P_n from here. S_0 minus S_2 divided by S_1 minus S_2 . What's interesting observation here, well, first of all, that we can find this risk-neutral probabilities. And we actually could price our...

37:07 — it will be sum of all the smaller steps. So, what's expected value of W_n ? Well, it is sum of the expected values of all dW s, and each of them is 0 , so it is 0 . It's, in fact, martingale, it stays 0 . Well, what is the standard deviation, or variance, of our dW_n ? Well, it will be sum of squared of each step. Yeah, and what I forgot to say-- to tell here, is that probability of us going up-- we are going by...

52:10 — we'll use our-- either Ito's formula. Well, let's do that. So we continue to differentiate that, and we get df minus $a dS$. And our f is stochastic, we use our Ito's formula, would be d of dt times dt plus $df dS$ times dS plus our stochastic term, plus $1/2 d$

2 f by dS squared dS squared. And we don't forget to subtract this. And if we just continue further and start collecting terms-- so we'll expand our dS according to our formula....

63:50 — gives you the heat equation, and this is very well-studied and solved problem, even though it has to be numerical. So as any partial differential equation, to solve it, it will require the initial-- or actually, in this case, final conditions and boundary conditions. And what is our final condition? Well, our final condition is actually our final payoff, which will be this. This is our final condition. This we know. And what will be our boundary condition for call? Will be that at 0,...

75:26 — which you have at hand, or at least approximate the payout, because if you did it, you know the price now. Or you know arbitrage. The other way to do it is form some complicated portfolio and find misplacement. And then it becomes arbitrage. Nowadays, it's practically impossible, but people did a lot of things with more complicated options even 15, 20 years-- well, 25 years ago. I think Jake has a good story about that, how he was starting in FX, options. Yeah, just...

4. Câu/ý tiêu biểu từ transcript

“So then the variance here will be sum of expected values of dW_i squared, each of them will be 1 with probability 1/2.”

“So basically, what I'm trying to say is that, if we put our steps on a period of time, that's what's our variance.”

“But for our risk-neutral assumptions, remember-- so we said that in discrete case, the whole point of risk-neutral probability was that expected value of our stock should be-- our stock may be compounded by interest rate.”

“Again, some simple examples and derivatives which we will be looking at will be call/put options and forward contracts.”

“So we basically form a portfolio of option, of full value of option, and some stock such the combination is riskless.”

“So basically, by holding option and some amount of stock, this amount of stock, our portfolio becomes riskless, which is very important practical conclusion.”

“And so our P-- and I'll call it neutral risk-- neutral probability, S_1 plus 1 minus P_n S_2 .”

“Well, I also plotted here implied volatility for-- I think I did it for call here.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 20: Building the First Federally (CFTC) Regulated Exchange Dedicated to Trading on Events

Source transcript: mit_18_642_transcripts\19_8XrYjnDHmE4.json

1. Chủ đề nhận diện trong video

machine learning

black-scholes/options

event markets

portfolio/risk

volatility

2. Giảng lại đầy đủ ý chính

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Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:18 — PETER KEMPTHORNE: OK. Tarek has just joined us, I believe. TAREK MANSOUR: Hey, everyone. PETER KEMPTHORNE: Hey, Tarek. Well, welcome. And just as a brief introduction, I'm really very proud to introduce Tarek Mansour to everyone. He graduated from MIT about six years ago or seven years ago, and he was actually in this class. And so we're really very pleased that he has come as a guest lecturer. And his background, he left-- when he left MIT, he joined really the giants of Wall...

06:47 — There wasn't a clear answer. Like, OK, why? They're like, well, because of they could be manipulated. I'm like, OK, but that's solvable, same way we solved the stock market. And then the other one is, well, it could be gambling. It's like, well, why is it gambling and something like a grain future is not? I mean, some people are betting on grain futures. And sure, some people are hedging on grain futures, but it's the same thing for Brexit. A lot of people...

12:38 — You negotiate bilaterally with the investment bank instead of just basically doing it on exchange, where there's other people that are independently pricing the events so that you can basically know what the right price is to basically make the transaction at. I think there are a lot of similarities. And frankly, I haven't-- we don't really discern them quite much. I think over time the two are converging, and they're going to end up being one and the same. The fundamental difference, I would...

18:39 — front for their positions. And then we basically mark to market every day. Actually, we mark to market on a real-time basis, so we do it in real time. We are going to introduce leverage next year, I think, and that's going to be pretty interesting. We're thinking about what our margin model is going to be. And maybe for the class, I'll explain. So in a lot of financial markets, you-- what people consider margin leverage is when you buy-- when hedge funds basically...

24:49 — And we're seeing that. We're seeing that. VASILY STRELA: Right. Very interesting. Thank you. TAREK MANSOUR: Steven. STUDENT: Yeah, I have a question about-- because you said that you were trying to also get into sports as well. So I was wondering if you planned on offering-- or similar things to other sports books and stuff, like traditional bets, like money lines and stuff like that, or would it be more of the work, more like a broader prediction-type sports bets? TAREK MANSOUR: That's a...

30:44 — being able to make it. Because to me, it seems like even if I had an idea of, oh, this would be a great thing to turn into an exchange, just the idea of actually making that happen seems really hard to achieve. From the idea to actually getting it into a reasonable product, how did you-- what did you have to do? Did you just have to literally start emailing people in the industry, and then start working your way up? How did that...

36:11 — not actually possible? Or did you just never have that in your mind? You just kept going. TAREK MANSOUR: Not really, basically. I mean, definitely not a people-driven thing. What I've learned is, also, a lot is like, don't really listen to authority figures. You'd be surprised by the extent to which people just don't really know what they're doing. And by people, I mean not average. I'm talking the President of the United States, the Supreme Court Judge, Elon Musk, like Mark Zuckerberg. People...

42:32 — Maybe Jim Simons is an exception, but they're not exactly the smartest. LTCM blew up, and those were the smartest guys. A lot of the others, like Steve Cohen and some of the others, they're pretty smart, but they're more well-rounded. They're just much more well-rounded than just being the best at math. And I think there's a lot of learning to that. JAKE XIA: Great. Thank you. TAREK MANSOUR: Well, great. Well, thanks. Thanks so much for having me, everyone. And hopefully we'll catch...

4. Câu/ý tiêu biểu từ transcript

"You negotiate bilaterally with the investment bank instead of just basically doing it on exchange, where there's other people that are independently pricing the events so that you can basically know what the right price is to basically make the transaction at."

"We do have some products where, for example, if there's a lot of strong correlation between two products, we return some collateral to the participant."

"So when you go to a casino and you roll a dice, that's a risk that you're creating for the purpose of entertainment, whereas Brexit, that's not a risk that you're creating for the purpose of entertainment."

"So we are fully cash collateralized right now, so what it means is people put the money up front for their positions."

"So let me hand it over to Tarek to share his experiences and details on Kalshi."

"So in a lot of financial markets, you-- what people consider margin leverage is when you buy-- when hedge funds basically buy, let's say, interest rate futures or S&P future."

"Kalshi is a simpler model where we actually ask for 100% upfront."

"And so people can basically do what they want with that risk, whether they want to get exposed to it more by speculating, or they want to expose-- get exposed to it less by hedging."

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 23: Introduction to Machine Learning

Source transcript: mit_18_642_transcripts\20_kTsiell_YBA.json

1. Chủ đề nhận diện trong video

machine learning

black-scholes/options

volatility

portfolio/risk

probability

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Volatility modeling cho thấy biến động thay đổi theo thời gian và ảnh hưởng trực tiếp đến risk, sizing và option/CW. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying. ML chỉ nên dùng sau khi feature, label và validation sạch; trong tài chính, overfit là rủi ro trung tâm.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:16 — PROFESSOR: But yeah, we are very glad to have John Hull with us here. So John, please. JOHN HULL: Well, thanks very much. It's a real pleasure to be here. Let me just say a little bit by way of introduction. I probably got a similar background to many of you. Math was always my best subject at high school. That was in the UK. I, not surprisingly, ended up doing a math undergraduate at Cambridge. And after graduating, I decided that I wanted to...

08:57 — And there are certain rules they use to decide whether a loan should be accepted or not. And those rules have been worked out. Then we could automate their activities. And that would just be the third Industrial Revolution. If we didn't know the rules, we could actually use machine learning to determine the rules. We'd collect lots and lots of data on loans that came before the loan officers and the decisions that were made. And it would be an application of machine learning....

17:59 — seems to be the best model for explaining the data. Typically, you can find a really complex model, sometimes anyway, that will explain the data in the training set but then it does not do well on the validation set. So the model can be too complex. So we'll see. So you want to, typically, you want to-- a rule of thumb is you want to increase model complexity until the model starts to perform worse on the validation set. So once you've chosen a...

27:25 — this activation function to get $V_2,1$. And then so what are the things-- what are the parameters of this model? The parameters are the weights. What weights do we apply to the V 's in going from one layer to the next layer? OK. And there's a lot of weights in this example that you got in front of you there because when we're determining $V_1,1$, we've got a weight associated with $V_0,1$, $V_0,2$, $V_0,3$, and so on. But even if there are only three inputs,...

36:57 — But clearly, if we could just go back to the previous slide a moment, I think you can see here, if you choose a really small learning rate, then the gradient at 1 would be the same but you'd move only a very short amount, and then you calculate the gradient

again, and move a very short amount, and so on. You'd never actually reach the bottom of the-- you'd never in a reasonable time reach the bottom of the curve if you made the...

45:48 — is we've looked at the relationship between volatilities and how volatilities change as the return on the asset changes. The standard way of doing hedging, of course, is to say, well, I mean, let's talk about delta hedging, but it applies to other hedging as well. With delta hedging, you say, well, how does the price of the option change when the price of the underlying asset changes? And we'll assume that the implied volatility remains the same. That's the standard assumption. Implied volatility remains...

55:10 — So suppose we've got two players. So I'm playing against you. And there's a pile of matches on the table. And we take turns. Each of us can pick up one, two, or three matches. And you lose the game if you have to pick up the last match. OK. So pile of matches. Take turns picking up one, two, or three matches. And the person who has to pick up the last match loses. Well, if you are being good mathematicians, you'd work out...

63:34 — because, remember, your opponent's behaving randomly. And when you've learnt a lot, you can behave much better than your opponent. I've done other tests where the opponent is learning as well, in which case picking up two matches or one match is going to allow your opponent to win. And so you have negative numbers opposite the 2 and the 1. Next slide, please. This Nim game illustrates, admittedly in a fairly simple situation, how reinforcement learning works. Very often you've got a much more...

73:27 — and then use the results of that to add a third layer, or is the three layer neural network just built independently? JOHN HULL: That's a good question. As far as I know, you've got to start from scratch when you go from, say, two layers to three layers. You can't use what you've done with two layers as a shortcut to doing three layers. I may be wrong there, but I don't think there's any way. PROFESSOR: Right. Actually, for the class's benefit, I...

4. Câu/ý tiêu biểu từ transcript

“But basically, I generated 10,000 call option prices using the Black-Scholes-Merton model, and added a normally distributed error.”

“Typically, you can find a really complex model, sometimes anyway, that will explain the data in the training set but then it does not do well on the validation set.”

“So we calculated a neural network to investigate this and to see what's the relationship between the daily asset price return and the change in the implied volatility.”

“So if the variables in the first layer were a linear function of the input layer, the variables in the second layer were linear functions of the variables in the first layer and so on, and you went all the way through, then it would just be a linear model.”

“Say, well, this model seems to do better than this other model when tested on-- the parameters of the model are determined using the training set.”

“Or CVaR95 is where you want the conditional value at risk, which is the average beyond the 95th percentile of the loss distribution to be as low as possible.”

“And there's a certain probability that you'll choose the decisions that actually works out best so far, which is called exploitation.”

“And so, typically, you put about 60% of your data into the training set, 20% into the validation set, and 20% into the test set.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.

- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 24: Stochastic Calculus

Source transcript: mit_18_642_transcripts\21_5cruqmIF6I0.json

1. Chủ đề nhận diện trong video

probability

stochastic calculus

stochastic processes

black-scholes/options

portfolio/risk

2. Giảng lại đầy đủ ý chính

Bài này trong MIT 18.642 tập trung vào việc xây một mảnh ghép của tư duy toán tài chính. Khi đọc transcript, nên chú ý cách giảng viên đi từ khái niệm nền, đặt giả định, rồi dùng mô hình để giải thích sản phẩm hoặc hiện tượng tài chính. Phần xác suất nhắc rằng kết quả tài chính là phân phối khả năng, không phải một con số chắc chắn. Phần quá trình ngẫu nhiên xem giá là đường đi theo thời gian, nên drawdown, hitting time và thứ tự dữ liệu rất quan trọng. Phần danh mục/risk nhấn mạnh rằng chọn mã tốt chưa đủ; correlation, concentration và sizing quyết định rủi ro thực tế. Black-Scholes và risk-neutral valuation là nền để hiểu option/CW: thời gian, volatility và moneyness quan trọng không kém hướng đi của underlying.

Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:10 — PETER KEMPTHORNE: All right. So today's topic is stochastic calculus. And this topic is really an engaging one in quantitative finance because it extends ordinary calculus with real variables to calculus that depends upon stochastic processes, and in particular, Brownian motion processes. And so when we are thinking about how to model the value of a stock or of a derivative, such as a simple option or a more complex option, we can model the dynamics of these assets or derivative contracts as integrals of...

11:20 — Because, obviously, the first one is like a Riemann integral. What about the second one? Is there an actual formal definition of-- I guess it's an Ito integral. PETER KEMPTHORNE: Right. And we will get there in a moment. But if we're integrating from-- well, here it's from 0 to t . If we're integrating the process over increments, we're basically going to be summing over the increments, although summing on the infinitesimal timescale over the entire period between 0 and t . And so if we...

24:04 — i minus 1 T over n -- the integral from i minus 1 T over n to i times T over n dBs. So here, I'm thinking of basically dividing this increment from 0 to T into different increments of width capital T over n . And if we approximate-- or each of these integrals here are basically looking at the integral of or the change in B_s from these lower limits to the upper limits. And so the integral of dBs of ω , in a consistent...

36:14 — AUDIENCE: Yeah. Oh, yeah. But what if a_i is not independent? Because when you take the expectations. PETER KEMPTHORNE: OK. Yep. Yep. Let's see. OK, a_i ω is an element of $\mathcal{F}_{t_i}$. So it's constant on $\mathcal{F}_{t_i}$. So let's see. So a_i of ω and $B_{t_i} + 1$. This increment into the future from time t_i is independent of the past. And so its expectation, if we take the product of these two and take expectations, we can take the...

48:43 — obvious. What's not necessarily obvious, though, is that our process X_t is a martingale with respect to this stochastic process model given by the Brownian motions. So if we look at what it means for a process to be a martingale, if we look at the conditional expectation of the time t plus k value given time t , then it equals the value of the process at time t , little t , plus the integral of the process from time t to t plus k . And...

60:43 — And if we take the limits of those, we get the integral of f' dB s plus $1/2$ the integral of the second derivative. So Taylor series gives us this formula. And we can think of developing a notation with Ito calculus, which is that our differential of f at B s is equal to f' of B s dB s plus $1/2 f''$ times B s times the squared of dB , which is equal to ds , the increment of time. So this is our Ito's...

71:56 — is that the variance of the function, f of T , B_T , which is bounded by the squared expectation, is finite. And so this will be finite, the variance, if that second term has finite variance. And so for that to have finite variance, the variance is the integral of the square of that argument df/dx , with respect to time. Now, what's interesting with this Ito formula is we can apply it to a function little f , which is the exponential of σx minus...

4. Câu/ý tiêu biểu từ transcript

“And a standard Brownian motion model is one where we have our process X , depending upon the underlying Brownian motion-- standardized Brownian motion B of t , with a drift parameter μ , and a volatility parameter σ , or variance parameter σ^2 .”

“And if we consider, I guess, the covariances of the X process given by these Ito integrals from 0 to little t , then these will-- or this X_t will be a mean 0 Gaussian process with independent increments.”

“With Brownian motion with drift, the ruin problem, we can actually use some Ito-process ideas to deal with this problem, where we assume that X_t is a Brownian motion with drift μ and variance parameter σ^2 .”

“What's not necessarily obvious, though, is that our process X_t is a martingale with respect to this stochastic process model given by the Brownian motions.”

“So this little f evaluated at x equal to B_t -- so it's a transformation of the Brownian motion process paths, B_t by this function-- is actually a scaled lognormal random variable.”

“And this topic is really an engaging one in quantitative finance because it extends ordinary calculus with real variables to calculus that depends upon stochastic processes, and in particular, Brownian motion processes.”

“And for Brownian motion with drift, this increment is exactly distributed as a normal random variable, with that mean and variance.”

“So with Brownian motion and drift, we have the notation for a stochastic differential equation, which is simply the increment of X is equal to $\mu \Delta t$, or dt rather, plus σdB of t .”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
- Phân biệt mô hình giải thích quá khứ với mô hình dự báo tương lai.
- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.

Lecture 25: Stochastic Calculus (cont.); Stochastic Differential Equations

Source transcript: mit_18_642_transcripts\22_H4V29wkHYb4.json

1. Chủ đề nhận diện trong video

stochastic processes

stochastic calculus

black-scholes/options

probability

portfolio/risk

2. Giảng lại đầy đủ ý chính

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Đọc video này nên theo mạch: giảng viên giới thiệu vấn đề, xây khái niệm, đưa công cụ toán hoặc mô hình, rồi nối sang ứng dụng tài chính. Các ví dụ và thuật ngữ trong transcript được giữ theo đúng tinh thần bài giảng thay vì biến thành khuyến nghị đầu tư.

3. Timeline nội dung theo transcript

00:11 — PETER KEMPTHORNE: And so from last time, we covered the Itô formula for the single case where we have the integral-- or sorry, the function of a Brownian motion, function of one variable. And importantly, with this function of a one variable, we get that the differential of the function is equal to the first derivative of the function evaluated at the Brownian motion plus $1/2$ the second derivative of the function. And so this is Itô's formula. And when we apply Itô's formula in...

10:51 — Now, that 1.1618 says that we expect the price to increase by about 16%. And if we think of a half year and a drift rate of 0.3, we might have thought it would be 15% if we didn't follow through the details. So it's a bit higher return. And similarly, for the volatility or standard deviation of this price, it turns out that the volatility or variance of the price at time capital T is 0.335 squared times the square of P_t . So the...

21:02 — is a mean growth rate times Δt plus $\sigma P_t \Delta B_t$. So there's a random term and a deterministic term given time little t . And if we do the same for the increment of the derivative, we then basically substitute into dG , Δt 's and ΔB 's. So we get the same constant factor times Δt plus $dG dP$ times $\sigma P_t \Delta B_t$. And the brilliant argument for solving for the option price or derivative price is to consider constructing a...

31:58 — how should that be valued? Great. OK. So what's important with this Black-Scholes partial differential equation is that there are different boundary conditions that vary by the derivative that we are working with. And so with call options, we have basically the final value at time capital T. The maturity of the option is given by the hockey stick function of P_t minus K if that's greater than 0 or 0. And similarly, for the put option, it has this terminal value. Now, what's, as...

43:49 — of this function at time 0. And if we integrate our u_0 function with the Dirac delta function evaluated at x minus s , then we simply get back this u_0 function at x . So the Dirac delta function allows us to pull out in an integral the u_0 function itself. So there's an

identity relationship between u_0 and this transform of u_0 with the Dirac delta function. Now, if we consider different solutions. We can index solutions to this heat equation by δ -- or by...

55:41 — and rescaling the space. So we must have that u of xt is equal to u of $\sqrt{t}$ if we consider this particular transformation. Now, with the choice of α and β , those are arbitrary. We can choose α equal to $1/\sqrt{t}$. And then that means our u of xt is equal to u of $x/\sqrt{t}$ and 1 , which is a function just of $x/\sqrt{t}$. So we have a transformation of our...

66:01 — So what we have here are conditions for the drift not getting too wild and the volatility not getting too wild in order for there to be solutions to this. A really important theorem that we won't get into but which are dealt with in advanced books on stochastic calculus deal with, the existence and uniqueness of solutions to these stochastic differential equations. And so if we consider two processes, x_t and y_t , that are both solutions to the stochastic differential equation, then subject to...

76:35 — So if we just tentatively assume this possible solution, we can apply the chain rule to get this first equation. dx_t is the derivative of x times the bracketed term, then plus $\frac{1}{2} \sigma^2$ times the derivative of the right-hand term with respect to t . And we can then use coefficient matching as per these expressions. And at the end of the day, we end up getting our Ornstein-Uhlenbeck process. If we start at x_0 , it-- well, let's see. If we consider the process...

4. Câu/ý tiêu biểu từ transcript

“And then let's see, if we consider, I guess, looking at what the conditional distribution is of $\log P_t$, it's given by this Brownian motion, which gives normally distributed log values with a particular mean μ^* and variance σ^* .”

“And if we-- I guess, before doing this, I'd like everyone to just think about how is this set of paths going to transform when we take the exponential of them, so basically the exponential of Brownian motion with constant drift and variance?”

“So with geometric Brownian motion, we have this process that has a mean level that's growing arbitrarily large, but the process is, with probability approaching 1, approaching 0 for all paths.”

“So this is an important property of geometric Brownian motion, that the growth of the process basically grows at a constant rate with some variability.”

“So if we have this formula for the increment, then we basically have a constant term plus an integral of Brownian motion, which has mean 0 as a random variable.”

“And what's interesting about this process is that it has a limiting distribution that's not like a non-stationary random walk of Brownian motion.”

“And so log price is going to follow a generalized Wiener process with a drift rate given by $\mu - \frac{1}{2}\sigma^2$ and variance rate equal to σ^2 .”

“So we get that-- we basically get our solution for geometric Brownian motion coming out from that differential equation.”

5. Điểm cần ghi nhớ

- Hiểu khái niệm và giả định trước khi dùng công thức.
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- Trong tài chính, phải luôn hỏi rủi ro, xác suất, chi phí và điều kiện mô hình sai.